ASK RED BEAR ASSISTIVE AI DEVELOPMENT STATUS & COMPLETION ASSESSMENT

Comprehensive Technical Audit of Assistive AI Architecture, Speech Pipeline, Sovereign Governance, and Production Roadmap

Cowessess First Nation | Chief Red Bear Children's Lodge
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Governance Classification: Strictly Assistive AI — Human-in-the-Loop Guaranteed

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■■ MANDATORY AI ASSISTIVE & LEGAL NOTICE

Ask Red Bear is strictly assistive and advisory. It possesses zero legal authority to make autonomous child welfare screenings, custody determinations, or statutory child protection decisions. All outputs are generated as assistive drafts and require independent human review and legal verification by a qualified CRBCL caseworker or supervisor. Crucially, the backend completion engine currently runs on a simulated provider (FakeAiProvider); live cloud LLM integration requires CRBCL Board budget authorization and provider selection.

1. EXECUTIVE SUMMARY & ASSISTIVE SCOPE

Ask Red Bear is a custom-engineered artificial intelligence capability embedded within the CRBCL Family Wellness platform. Developed to alleviate heavy administrative burdens on frontline caseworkers, Ask Red Bear provides natural language record retrieval, schedule summarization, and hands-free voice transcription. In strict compliance with Indigenous OCAP® data principles and Canadian child welfare standards, the system is designed never to make autonomous decisions or replace human caseworkers.

2. CURRENT ARCHITECTURAL STATE & SIMULATION DISCLOSURE

• **LIVE VERIFIED (Backend RBAC & Logic):** The Authorization-First Context Gateway, active Case Restriction filters, deterministic tool routing (get_case_summary, get_my_cases, get_overdue_goals), field redactions, and database audit logging are 100% written, integrated, and verified with passing automated test suites (test_ask_red_bear_speech.py).

• **STUBBED / SIMULATED (LLM Generation):** The final model completion step in gateway.py currently utilizes FakeAiProvider (generating synthetic assistive summaries). No live API keys for external cloud LLMs (such as Anthropic Claude 3.5 Sonnet or OpenAI GPT-4o) have been stored in the production environment. This intentional design prevents unmonitored API billing until leadership authorizes an enterprise AI budget.

• **LIVE VERIFIED (Speech-to-Text Input Aid):** The browser-based speech dictation interface (SpeechDictationControl.jsx) and backend transcription endpoint (/api/v1/ask-red-bear/transcribe) function as an ephemeral input aid. In-memory audio recordings are transcribed directly into the text box for user review and editing before prompt dispatch.

3. AUTHORIZATION-FIRST SECURITY & FIELD REDACTION ENGINE

Ask Red Bear implements rigorous defense-in-depth protections to prevent unauthorized data exposure, hallucination, or prompt injection:

Security Control Layer	Implementation Mechanism	Child Welfare Protection Benefit
Prompt Guard Filter	inspect_prompt_safety() regex & semantic check	Rejects prompt injection, jailbreak attempts, or instructions to ignore system safety rules.
Case Restrictions	AiGateway.get_authorized_case_ids()	Completely excludes cases from retrieval context if the user has a registered family conflict of interest.
Field Redaction	Dynamic JSON stripping before prompt compilation	Automatically removes child medical notes, reporter identities, and financial accounts unless explicitly permitted.
Immutable AI Audit	AiRequestAudit database table logging	Logs user ID, prompt token counts, latency, and estimated cost in CAD for every single AI interaction.

4. KNOWLEDGE BASE & RETRIEVAL-AUGMENTED GENERATION (RAG)

• **Cowessess Miyo-Pimatisiwin Act Ingestion:** Chunking, indexing, and embedding the full statutory text of the Cowessess Child and Family Wellness legislation so Ask Red Bear can cite specific legal sections.

• **CRBCL Operational Policy Ingestion:** Vectorizing internal manuals regarding customary care protocols, travel mileage reimbursements, and emergency after-hours intake procedures.

• **Sovereign Vector Storage:** Enabling pgvector within the existing PostgreSQL database to ensure all vector embeddings remain strictly within CRBCL's sovereign boundary.

5. PRODUCTION PROVIDER COMPARISON & COST PROJECTIONS

Provider Option	Data Sovereignty & Privacy	Estimated Monthly Cost	Recommendation
Option A: Anthropic Claude 3.5 Sonnet (AWS Canada)	Data remains within Canada (ca-central-1); zero model training on customer data.	$150–$300 CAD / mo (~10,000 queries)	**RECOMMENDED:** Highest reasoning, strong safety guardrails, low cost.
Option B: Azure OpenAI (Canada East)	HIPAA/SOC2 certified; isolated Canadian enterprise tenant.	$200–$400 CAD / mo	**VIABLE ALTERNATIVE:** Seamless if CRBCL adopts Microsoft 365.
Option C: Self-Hosted Llama-3 (On-Premise)	100% on-premise Cowessess data sovereignty; zero third-party transmission.	$600–$1,200 CAD / mo (dedicated GPU)	**HIGH SECURITY OPTION:** Ideal long-term for complete sovereign isolation.

6. OPERATIONAL COMPLETION ROADMAP

• **Weeks 1–2:** Board approval of AI budget and signing of enterprise sovereign agreement with Anthropic AWS Canada.

• **Weeks 3–4:** Ingest Cowessess Miyo-Pimatisiwin Act into pgvector database; conduct prompt testing with senior caseworkers.

• **Weeks 5–6:** Deploy live API keys, establish automated monthly token cost alerts, and conduct mandatory caseworker AI ethics training.